Definition. Given a function f defined on any interval around a, but not necessarily at a itself, we say that $\lim_{x\to a} f(x) = L$ if for any $\varepsilon > 0$, there is a $\delta > 0$ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

Problem 1. In this problem, we will show that $\lim_{x\to 2}(-2x + 1) = -3$, using the definition of the limit.

(a) Write down what the definition of the limit becomes when specialized to this particular case.

(b) Rewrite the inequality $|-2x + 1 - (-3)| < \varepsilon$ as an inequality involving $|x - 2|$.

(c) Based on part (b), given $\varepsilon > 0$, find a δ in terms of ε that works in the definition of the limit.

Solution. (a) For any $\varepsilon > 0$, there is a $\delta > 0$ such that if $0 < |x - 2| < \delta$, then $|-2x + 1 - (-3)| < \varepsilon$.

(b) $|-2x + 1 - (-3)|$ can be simplified to $|-2x + 4| = |(-2)(x - 2)| = 2|x - 2|$, so the inequality $|-2x + 1 - (-3)| < \varepsilon$ is equivalent to $|x - 2| < \varepsilon/2$.

(c) Take $\delta = \varepsilon/2$. Then if $|x - 2| < \delta$, we can run the calculation in (b) backwards to get

$$2|x - 2| < \varepsilon$$

$$|(-2)(x - 2)| < \varepsilon$$

$$|-2x + 1 - (-3)| < \varepsilon$$

This means that this δ works in the definition of the limit.

Problem 2. In this problem, we will verify that for any numbers a, b, c, we have $\lim_{x\to c}(ax + b) = ac + b$.

(a) Write down what the definition of the limit becomes when specialized to this particular case.

(b) Show that if $a \neq 0$, then given any $\varepsilon > 0$, taking $\delta = \varepsilon/|a|$ works.

(c) Show that if $a = 0$, then for any $\varepsilon > 0$, any δ works.

Solution. (a) For any $\varepsilon > 0$ there is a $\delta > 0$ such that if $0 < |x - c| < \delta$, then $|ax + b - (ac + b)| < \varepsilon$.

(b) First of all, we can rewrite $|ax + b - (ac + b)|$ as $|ax - ac| = |a| \cdot |x - c|$. If we take $\delta = \varepsilon/|a|$, then for any x satisfying $|x - c| < \delta$, we have

$$|ax + c - (ac + b)| = |a| \cdot |x - c| < |a| \cdot (\varepsilon/|a|) = \varepsilon$$

This means that this choice of δ works.

(c) In the case that $a = 0$, $|ax + b - (ac + b)| = |b - b| = 0$ regardless of the value x. Therefore, given any $\varepsilon > 0$, if we choose any $\delta > 0$, then for x satisfying $0 < |x - c| < \delta$, we have

$$|ax + b - (ac + b)| = 0 < \varepsilon$$

We conclude that any choice of $\delta > 0$ works.

Problem 3 (Optional)**.** In this problem we will consider the form of the definition of the limit. Consider the following similar sounding statement:

There is a $\delta > 0$ such that for all $\varepsilon > 0$, if $0 < |x - a| < \delta$, then $|f(x) - L| < \varepsilon$.

Argue that this means that f is constant and equal to L on some interval around a (except maybe a itself). Thus this statement is much stronger than what the definition of the limit asserts.[1]

Solution. Suppose that there is a $\delta > 0$ such that for all $x > 0$, if $0 < |x-a| < \delta$, then $|f(x)-L| < \varepsilon$. I claim that f must be constant and equal to L on the intervals $(a-\delta, a) \cup (a, a+\delta)$, in other words the interval $(a-\delta, a+\delta)$ with a removed. To show this, we will argue by contradiction, assuming that the conclusion we want is false and deriving a contradiction from this assumption. So assume that f is not constant and equal to L on these intervals. Then there is a point $x_0 \in (a-\delta, a+\delta)$, $x_0 \neq a$, such that $f(x_0) \neq L$. The distance $|f(x_0) - L|$ is then positive. Let's take $\varepsilon = |f(x_0) - L|$. The point x_0 satisfies $0 < |x_0 - a| < \delta$, so by our assumption, we should have $|f(x_0) - L| < \varepsilon$, but this reads $\varepsilon < \varepsilon$, a contradiction.

From the contradiction, we can conclude that our assumption was false, or in other words, that f is constant and equal to L on $(a-\delta, a) \cup (a, a+\delta)$.

Definition. Let f be a function defined on an interval around a. We say that f is continuous at a if $\lim_{x \to a} f(x) = f(a)$.

Problem 4. In this problem, we will show that the function $f(x) = x^3$ is continuous at 0.

(a) Write down what it means for this function to be continuous at 0 in terms of a limit.

(b) Write down what it means for this function to be continuous at 0 in terms of ε's and δ's.

(c) Given an $\varepsilon > 0$, explain how to choose an appropriate δ to verify the statement in part (b).

Solution. (a) $\lim_{x \to 0} x^3 = 0$.

(b) Two answers are possible here. One uses the ε, δ definition of the limit in (a): For any $\varepsilon > 0$, there is a $\delta > 0$ such that if $0 < |x - 0| < \delta$, then $|x^3 - 0| < \varepsilon$. The other uses the definition of continuity I gave in class: For an $\varepsilon > 0$, there is a $\delta > 0$ such that if $|x - 0| < \delta$, then $|x^3 - 0| < \varepsilon$. The difference is that that second one allows $x = 0$; in this case $|x^3 - 0| = 0 < \varepsilon$, for any positive ε, so the two definitions are equivalent.

(c) First of all, $|x - 0| = |x|$ and $|x^3 - 0| = |x^3| = |x|^3$, so given $\varepsilon > 0$, we need to choose a $\delta > 0$ such that if $|x| < \delta$, then $|x|^3 < \varepsilon$. If we take $\delta = \varepsilon^{1/3}$, this holds.

Problem 5 (Optional)**.** Argue from the ε, δ definition of continuity that if f is continuous at a and g is continuous at $f(a)$, then the composite function $h = g \circ f$, defined by $h(x) = g(f(x))$, is continuous at a.[2]

Solution. To show that h is continuous at a, we need to show that for any $\varepsilon > 0$, there is a $\delta > 0$ such that if $|x - a| < \delta$, then $|g(f(x)) - g(f(a))| < \varepsilon$. Given $\varepsilon > 0$, the continuity of g at $f(a)$ tells us that there is an $\delta_1 > 0$ such that if $|y - f(a)| < \delta_1$, then $|g(y) - g(f(a))| < \varepsilon$. Taking $\varepsilon_2 = \delta_1 > 0$, the continuity of f at a tells us that there is a $\delta_2 > 0$ such that if $|x - a| < \delta_2$, then $|f(x) - f(a)| < \varepsilon_2$. For h, we will choose $\delta = \delta_2$. Then if $|x - a| < \delta$, we have $|f(x) - f(a)| < \varepsilon_2 = \delta_1$, and so $|g(f(x)) - g(f(a))| < \varepsilon$, as required.

Problem 6. Consider the following function

$$f(x) = \frac{2x^{4/3} - x\sin(x)}{\sqrt{x^2 - 1}}$$

[1] Expressions of the form "for all x" and "there is a y" are called *quantifiers*. In this exercise we see that these quantifiers cannot be permuted without changing the meaning of a statement.

[2] Hint: Given $\varepsilon > 0$, take $\delta_1 > 0$ from the continuity of g, then use $\varepsilon_2 = \delta_1$ and the continuity of f to get a δ_2.

(a) Where is f defined? Write the answer in terms of intervals.

(b) Evaluate the derivative f' of f. You should write out all the steps of computing the derivative, and indicate the differentiation rule you are using at each step.

Solution. (a) For the expression

$$\frac{2x^{4/3} - x\sin(x)}{\sqrt{x^2-1}}$$

to be defined, two things need to be true. First, we need that $x^2 - 1 \geq 0$, so that $\sqrt{x^2-1}$ is defined, and second, we need $\sqrt{x^2-1} \neq 0$. Together, these conditions are equivalent to $x^2 - 1 > 0$. Using the factorization $x^2 - 1 = (x+1)(x-1)$, we can see that this is true on $(-\infty, -1) \cup (1, \infty)$, as in the first interval, both factors are negative, and in the second, both are positive.

(b) The function f is a ratio of two functions $f = g/h$, where

$$g(x) = 2x^{4/3} - x\sin(x)$$

$$h(x) = \sqrt{x^2-1}$$

Let's first differentiate g and h. Using the power rule for the first term and the product rule for the second, we have

$$g'(x) = \frac{8}{3}x^{1/3} - \sin(x) - x\cos(x)$$

For h, we use the chain rule and the power rule to obtain

$$h'(x) = \frac{2x}{2\sqrt{x^2-1}} = \frac{x}{\sqrt{x^2-1}}$$

Now applying the quotient rule, we have

$$f'(x) = \frac{g'(x)h(x) - g(x)h'(x)}{h(x)^2} = \frac{\left(\frac{8}{3}x^{1/3} - \sin(x) - x\cos(x)\right)\sqrt{x^2-1} - \frac{(2x^{4/3} - x\sin(x))x}{\sqrt{x^2-1}}}{x^2-1}$$